Science practice booklet: Grade 11 expectations

Grade 11 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 11 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: HS-LS1-5, HS-LS1-6, HS-LS1-7, HS-LS2-3, HS-LS2-4, HS-LS2-5, HS-LS2-6, HS-LS2-7

A worked example

In a mangrove food chain, the producers capture 40000 kJ of energy from sunlight. The chain is: mangrove leaves and algae, then snails, then crabs, then herons. About 10 percent of the energy is passed on at each step. How much energy reaches the herons, and what happens to the rest?

1. There are three transfers: producers to snails, snails to crabs, crabs to herons.
2. Producers to snails: 10 percent of 40000 = 4000 kJ.
3. Snails to crabs: 10 percent of 4000 = 400 kJ.
4. Crabs to herons: 10 percent of 400 = 40 kJ.
5. The rest is used by each organism in cellular respiration to move, grow and stay alive, and most of it ends up as thermal energy released to the surroundings.

40 kJ reaches the herons; the rest is used in respiration or lost as heat.

Questions

1. Which balanced equation shows photosynthesis?

   A. C6H12O6 + 6O2 makes 6CO2 + 6H2O

   B. 6CO2 + 6H2O makes C6H12O6 + 12O2

   C. 6CO2 + 6H2O makes C6H12O6 + 6O2

   D. 6O2 + 6H2O makes C6H12O6 + 6CO2

2. Aerobic respiration is C6H12O6 + 6O2 making 6CO2 + 6H2O + energy. Count the oxygen atoms on each side of the equation. What does your answer show about matter?

   _____

3. In a mangrove bay near Abu Dhabi, the algae and mangrove leaves capture 60000 kJ of energy from sunlight in a day. About 10 percent is passed on at each step along the chain: producers, snails, crabs, herons. How much energy is available to the crabs?

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4. In a coastal reserve, a group of crabs (secondary consumers) needs 200 kJ of energy as food each day. About 10 percent is passed on at each step. How much energy must the producers capture each day to support the crabs? Why does this limit the number of top predators?

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5. Why does cellular respiration release energy overall?

   A. Forming the new bonds releases more energy than breaking the old bonds takes

   B. Breaking the old bonds releases more energy than forming the new bonds takes

   C. Energy is created when the atoms of glucose are destroyed

   D. Oxygen is a fuel that gives out its own stored energy

6. A mangrove tree makes glucose by photosynthesis, but it also needs proteins. Explain how the tree can make proteins from glucose.

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7. Which choice correctly names two processes that add carbon to the atmosphere and one that removes it?

   A. Adds: photosynthesis and burning fuel. Removes: respiration

   B. Adds: photosynthesis and decomposition. Removes: burning wood

   C. Adds: respiration and burning fuel. Removes: decomposition

   D. Adds: respiration and burning fossil fuels. Removes: photosynthesis

8. Mangrove forests store large amounts of carbon in their wood, roots and the mud around them. Explain why clearing a mangrove forest can add carbon dioxide to the atmosphere.

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9. A survey of a coastal area found 80 species of living things. Ten years later, after new building along the coast, a survey found 60 species. What percentage of the species was lost? Name one human activity that can reduce biodiversity.

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10. The Arabian oryx was once gone from the wild, but it was bred in protected reserves and then returned to its habitat. Which statement best explains how this helps to conserve a species?

    A. Once animals are bred, the species no longer needs its habitat

    B. Reserves protect habitat while breeding raises numbers for release

    C. Breeding makes the animals all alike, which keeps them safe

    D. Breeding does not change how many animals there are

11. In a lake, a fish that eats small plant-eating animals is removed by overfishing. What is most likely to happen next?

    A. Both the small animals and the algae increase

    B. The small animals decrease and the algae increase

    C. The small animals increase and the algae decrease

    D. Nothing changes because the fish did not eat algae

12. A town plans to build on a coast with a mangrove forest. Suggest a plan that reduces the impact on biodiversity, and name one criterion you would use to judge whether the plan works.

Section 2: Earth and space

Standards practised: HS-ESS2-4, HS-ESS2-6, HS-ESS3-1, HS-ESS3-2, HS-ESS3-3, HS-ESS3-4, HS-ESS3-5

A worked example

A reverse osmosis desalination plant produces 120000000 litres of fresh water a day. Suppose each person in a city uses 400 litres a day. How many people can the plant supply? Why does the process need a lot of energy?

1. Number of people = total water ÷ water per person.
2. 120000000 ÷ 400 = 300000 people.
3. In reverse osmosis, high-pressure pumps push seawater through a membrane that lets water through but holds back the salt.
4. Pumping at high pressure needs a lot of electrical energy.

300000 people; high pressure is needed to push the water through the membrane.

Questions

1. Which list has only renewable energy resources?

   A. Solar, natural gas and oil

   B. Solar, wind and tidal power

   C. Wind, oil and coal

   D. Natural gas, wind and coal

2. A solar park in the desert has a power output of 100 MW (100000 kW) for the equivalent of 6 hours of full sunshine a day. A home uses 30 kWh a day. About how many homes does the park supply in a day?

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3. Give one advantage and one disadvantage of generating electricity from natural gas, and one advantage and one disadvantage of generating it from solar panels.

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4. Which statement correctly describes how reverse osmosis makes fresh water from sea water?

   A. Sea water is boiled and the steam is collected

   B. Sea water passes through sand, which removes the salt

   C. Low pressure draws the salt through a membrane

   D. High pressure pushes water through a membrane that holds back the salt

5. A desalination plant takes in 1000 litres of sea water that contains 40 g of salt in every litre. It produces 400 litres of fresh water, and all the salt stays in the rest. How many litres of brine are left, and what is its salt concentration in g per litre (to the nearest whole number)?

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6. Burning fossil fuels moves carbon from underground stores into the air. What happens when the ocean absorbs some of this extra carbon dioxide?

   A. The sea water becomes more acidic, so shells and coral are harder to build

   B. The sea water becomes less acidic, so shells grow faster

   C. The sea water becomes saltier, so more fish can live in it

   D. Nothing changes, because gases do not dissolve in water

7. Explain how greenhouse gases such as carbon dioxide and methane warm Earth's surface. Why is some greenhouse effect necessary for life?

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8. Before industry grew, the air held about 280 parts per million of carbon dioxide. It now holds about 420 parts per million. What is the percentage increase?

   A. 33 percent

   B. 140 percent

   C. 50 percent

   D. 150 percent

9. As Arctic ice melts, dark ocean water is uncovered. Why can this make warming faster?

   A. Dark water reflects more sunlight than ice does

   B. Melting ice gives out heat that warms the air

   C. Ice absorbs most sunlight, so the water stays cool

   D. Dark water absorbs more sunlight, which melts more ice and uncovers more dark water

10. Why do scientists use computer models to forecast climate, and why do they run them with several different scenarios of future emissions? Give one result that low-lying coasts need to plan for.

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11. An underground aquifer below a farming area holds 800 million cubic metres of fresh water. People pump out 50 million cubic metres a year and only 10 million cubic metres a year flows back in. If this continues, in how many years will the aquifer be empty? Suggest one way to manage the water more sustainably.

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12. Suggest two ways a family can reduce its carbon dioxide emissions at home or when travelling. For each, explain how it reduces the amount of fossil fuel burned.

Section 3: Matter, forces and energy

Standards practised: HS-PS3-1, HS-PS3-2, HS-PS3-3, HS-PS3-4, HS-PS4-1, HS-PS4-3, HS-PS4-4, HS-PS4-5

A worked example

A student of mass 50 kg climbs a staircase that rises 4 m in 8 seconds. Use g = 10 N per kg. How much work does the student do against gravity, and what is the power?

1. Work = force × distance. The force needed to lift is the weight: 50 × 10 = 500 N.
2. Work = 500 N × 4 m = 2000 J.
3. Power = work ÷ time = 2000 J ÷ 8 s = 250 W.

2000 J of work; power 250 W.

Questions

1. A car of mass 1000 kg travels at 20 m per second. What is its kinetic energy? Use kinetic energy = ½ × mass × speed².

   A. 10000 J

   B. 200000 J

   C. 400000 J

   D. 20000 J

2. A 2 kg ball is held 20 m above the ground (use g = 10 N per kg). Find its gravitational potential energy. If it is dropped and air resistance is ignored, find its speed just before it lands.

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3. A motor takes in 500 J of electrical energy and gives out 400 J of useful movement energy. What is its efficiency?

   A. 80 percent

   B. 20 percent

   C. 125 percent

   D. 100 percent

4. Maryam's grandmother heats 2 kg of water for karak, from 20 °C to 70 °C. How much thermal energy is needed? Use Q = mass × specific heat capacity × temperature change, with a specific heat capacity of 4200 J per kg per °C.

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5. Which choice names the main way heat moves in each case? (1) A metal spoon in hot karak gets hot. (2) Warm air rises above a hot road. (3) You feel the warmth of the Sun on your face.

   A. 1 radiation, 2 conduction, 3 convection

   B. 1 convection, 2 conduction, 3 radiation

   C. 1 conduction, 2 convection, 3 radiation

   D. 1 conduction, 2 radiation, 3 convection

6. A farm pump lifts 1000 kg of water through 10 m in 100 seconds (use g = 10 N per kg). Find the work done and the power of the pump.

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7. A radio station broadcasts at a frequency of 100 MHz (100000000 Hz). Radio waves travel at 300000000 m per second. Use speed = frequency × wavelength to find the wavelength.

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8. Suppose the Hope Probe is 180000000 km from Earth. Radio waves travel at 300000 km per second. How long does a radio signal from the probe take to reach Earth?

   A. 600 minutes

   B. 6 minutes

   C. 60 minutes

   D. 10 minutes

9. A solar panel on a school roof has an area of 1.6 m² and receives 1000 W of sunlight on every square metre. Its efficiency is 20 percent. What is its electrical power output?

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10. Explain in terms of photons how a solar panel produces an electric current.

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11. Why can ultraviolet light damage skin cells when radio waves cannot?

    A. Each ultraviolet photon carries enough energy to break bonds in cell molecules

    B. Radio waves have a higher frequency, so skin blocks them

    C. There are always more ultraviolet photons than radio photons

    D. Ultraviolet waves travel faster than radio waves

12. A digital signal sends information as a code of 0s and 1s. How many different values can 3 bits represent, and how many can 8 bits represent? Give one reason why digital signals are used in phones and optical fibre cables.

Answer guide for parents

Section 1: Living things

1. C. $6CO_2 + 6H_2O$ makes $C_6H_{12}O_6 + 6O_2$. Carbon dioxide and water, with energy from light, make glucose and oxygen. There are 6 C, 12 H and 18 O atoms on each side. The light energy is stored as chemical energy in the bonds of glucose.
2. Left side: 6 (in glucose) + 12 (in $6O_2$) = 18 oxygen atoms. Right side: 12 (in $6CO_2$) + 6 (in $6H_2O$) = 18. The numbers are equal because atoms are rearranged, not created or destroyed (conservation of matter).
3. 60000 × 0.1 = 6000 kJ for the snails (primary consumers), then 6000 × 0.1 = 600 kJ for the crabs (secondary consumers).
4. 200 ÷ 0.1 = 2000 kJ for the primary consumers, then 2000 ÷ 0.1 = 20000 kJ for the producers. Each step up needs about ten times more energy below it, so only a small number of top predators can be supported.
5. A. Forming the new bonds releases more energy than breaking the old bonds takes. Energy is needed to break the bonds in glucose and oxygen, and more energy is released when the bonds in carbon dioxide and water form. The net release is used by cells to make ATP.
6. Glucose supplies carbon, hydrogen and oxygen. The tree takes in nitrogen (for example as nitrate dissolved in the water in the soil) and combines it with the atoms from glucose to build amino acids, which are joined into proteins.
7. D. Adds: respiration and burning fossil fuels. Removes: photosynthesis. Respiration, decomposition and burning put carbon dioxide into the air. Photosynthesis (and dissolving in the ocean) takes it out.
8. When the trees are removed, the wood and the carbon-rich mud can decompose or be burned. Respiration and decay by microbes then release the stored carbon as carbon dioxide, and fewer trees remain to take carbon dioxide in by photosynthesis.
9. Lost: 80 − 60 = 20 species. 20 ÷ 80 = 0.25, so 25 percent. Activities include habitat loss from building, pollution, overfishing and introduced species.
10. B. Reserves protect habitat while breeding raises numbers for release. Protected areas keep the habitat, food and water the species needs and limit hunting and disturbance. Breeding programmes increase numbers and keep a wide variety of genes, so animals can be released to rebuild the population.
11. C. The small animals increase and the algae decrease. Fewer small animals are eaten, so their number rises. They then eat more algae, so the algae fall. The species are linked in the food web, so removing one changes the others.
12. A possible plan: keep a protected strip of mangroves, build away from nursery areas, and plant new mangrove seedlings elsewhere. Criteria: the number of species counted before and after, the area of mangrove kept, or the water quality measured each year.

Section 2: Earth and space

1. B. Solar, wind and tidal power. The Sun, wind and tides are replaced continually. Natural gas, oil and coal formed over millions of years and are used up far faster than they form.

2. Energy = 100000 kW × 6 h = 600000 kWh. 600000 ÷ 30 = 20000 homes.
3. Natural gas: reliable at any time and can be switched on when needed; but it releases carbon dioxide and the supply is limited. Solar: no carbon dioxide while generating and the fuel is free; but output stops at night and varies with clouds and dust, so storage or back-up is needed and large areas of land are used.
4. D. High pressure pushes water through a membrane that holds back the salt. High pressure pushes sea water through a membrane that lets water pass but holds back most of the dissolved salt. The remaining brine is saltier and can harm sea life near the outlet if it is not mixed and spread out well.
5. Brine = 1000 − 400 = 600 litres. All the salt: 1000 × 40 = 40000 g. 40000 ÷ 600 = 66.7, so about 67 g per litre.
6. A. The sea water becomes more acidic, so shells and coral are harder to build. Carbon stored in the geosphere (coal, oil and gas) goes into the atmosphere as carbon dioxide. Carbon dioxide that dissolves in the ocean makes the water more acidic, which makes it harder for corals and shellfish to build shells and skeletons.
7. Earth's surface gives out infrared radiation. Greenhouse gases absorb some of it and send it out in all directions, including back down, so the lower atmosphere and surface stay warmer. Without any greenhouse effect, Earth would be far too cold for liquid water and most life.
8. C. 50 percent. Increase = 420 − 280 = 140. 140 ÷ 280 = 0.5, so 50 percent. Scientists know past amounts from air bubbles trapped in ice cores, and also from tree rings, sediments and coral records.
9. D. Dark water absorbs more sunlight, which melts more ice and uncovers more dark water. Ice reflects most sunlight, but dark water absorbs more of it. The extra energy warms the water, which melts more ice. This is a positive feedback, a change that makes itself bigger.
10. A model puts together many measured data and the physical laws for the air, ocean, ice and living things, so scientists can test what may happen over decades, which cannot be done in a laboratory. Different scenarios show how the results depend on human choices. Rising sea level (from melting land ice and the expansion of warmer water) is a result that low-lying coasts must plan for.
11. Net loss = 50 − 10 = 40 million cubic metres a year. 800 ÷ 40 = 20 years. Ways: use less water (efficient irrigation), reuse treated waste water, or use desalinated water to reduce pumping.
12. Examples: raise the air-conditioner setting a little and use efficient appliances, so the home needs less electricity from power stations that burn fuel; share car journeys or use the metro or a bus, so fewer litres of fuel are burned for each person.

Section 3: Matter, forces and energy

1. B. 200000 J. ½ × 1000 × 20^2 = ½ × 1000 × 400 = 200000 J. The speed must be squared, and the ½ must be used.
2. Potential energy = mgh = 2 × 10 × 20 = 400 J. All of it becomes kinetic energy: ½ × 2 × v^2 = 400, so v^2 = 400 and v = 20 m per second.
3. A. 80 percent. Efficiency = 400 ÷ 500 = 0.8, or 80 percent. The other 100 J is wasted, mostly as thermal energy and some sound, and spreads into the surroundings. Total

energy is still conserved.

4. Q = 2 × 4200 × (70 − 20) = 2 × 4200 × 50 = 420000 J
5. C. 1 conduction, 2 convection, 3 radiation. (1) Conduction through the metal. (2) Convection in the moving air. (3) Radiation, which crosses space. Heat always flows from the hotter object to the cooler one until they reach the same temperature.
6. Weight = 1000 × 10 = 10000 N. Work = 10000 × 10 = 100000 J. Power = 100000 ÷ 100 = 1000 W.
7. Wavelength = speed ÷ frequency = 300000000 ÷ 100000000 = 3 m
8. D. 10 minutes. 180000000 ÷ 300000 = 600 seconds, which is 10 minutes. Radio waves are electromagnetic waves, so they can cross empty space.
9. Input power = 1.6 × 1000 = 1600 W. Output = 20 percent of 1600 = 0.2 × 1600 = 320 W.
10. Light arrives as photons, packets of energy. When a photon is absorbed in the semiconductor (such as silicon), it gives its energy to an electron and frees it. The panel's structure pushes these freed electrons in one direction, which makes an electric current.
11. A. Each ultraviolet photon carries enough energy to break bonds in cell molecules. The energy of a photon rises with frequency. Ultraviolet photons have a much higher frequency than radio photons, so each one can break chemical bonds. A radio photon carries far too little energy, however many of them there are.
12. 3 bits: $2^3 = 8$ values. 8 bits: $2^8 = 256$ values. Digital signals are less affected by noise because the receiver only has to decide between 0 and 1, errors can be detected and corrected, and the signal can be copied and sent over long distances without loss of quality.